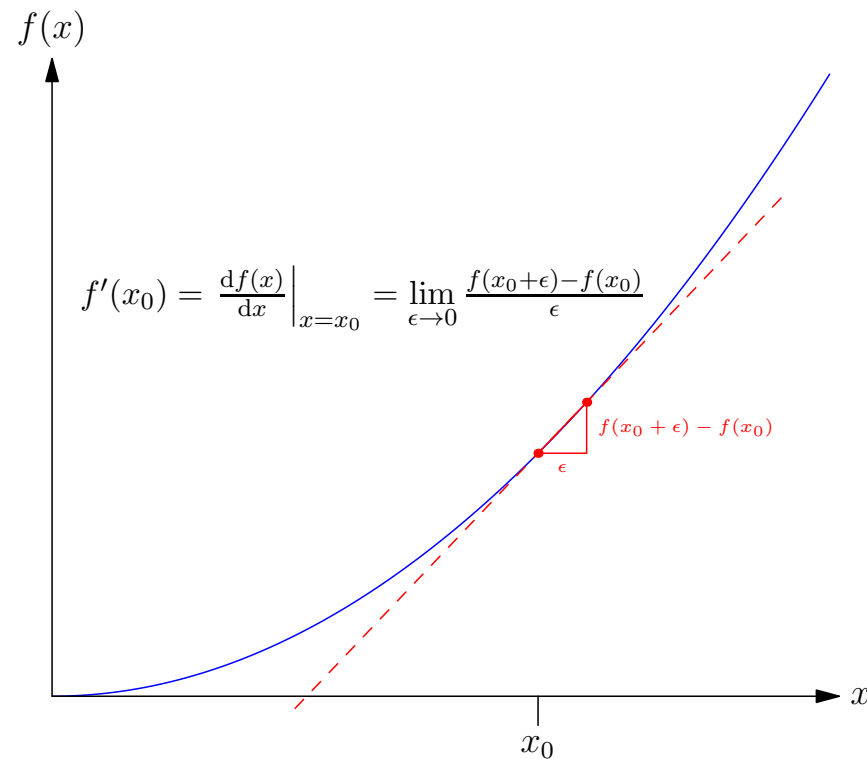


Mathematics of Machine Learning

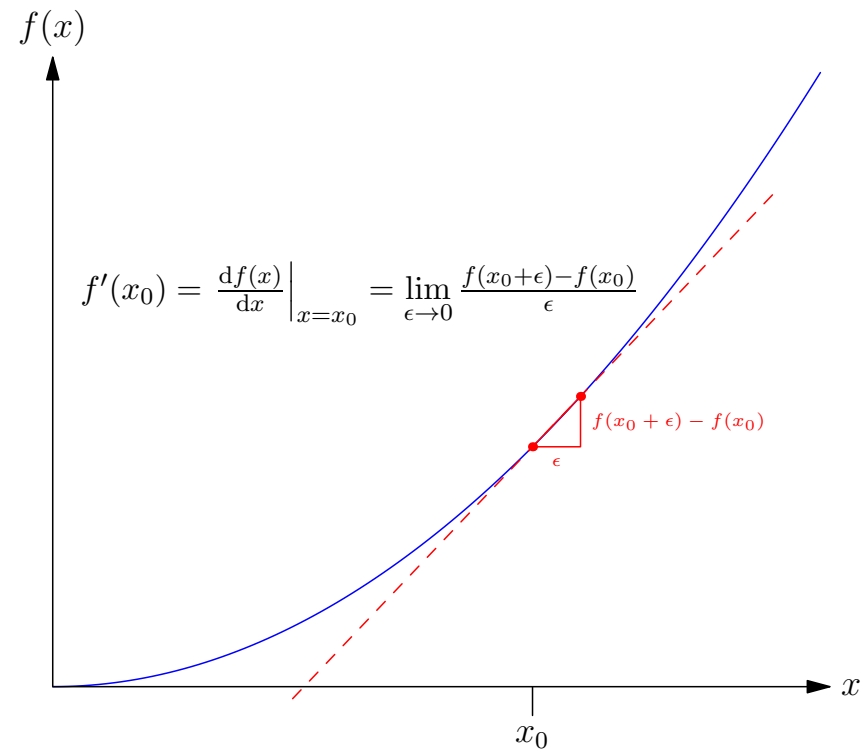
Differential Calculus



Differentiation, product and chain rules, vectors and matrices

Outline

1. **Why Calculus?**
2. Differentiation
3. Vector Calculus
4. Matrix Calculus



Why Calculus?

- Calculus is a fundamental tool of mathematical analysis
- In machine learning differentiation is fundamental tool in optimisation
- Integration is an essential tool in taking expectations over continuous distributions
- Both differentiation and integration crop up elsewhere

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Back to Basics

- You have all done A-level maths so should be familiar with the rules of calculus
- But, it is easy to forget the rules and sometimes we use quite sophisticated tricks
- Although the sophisticated tricks really speed up calculations, it pays to be able to understand where these tricks come from

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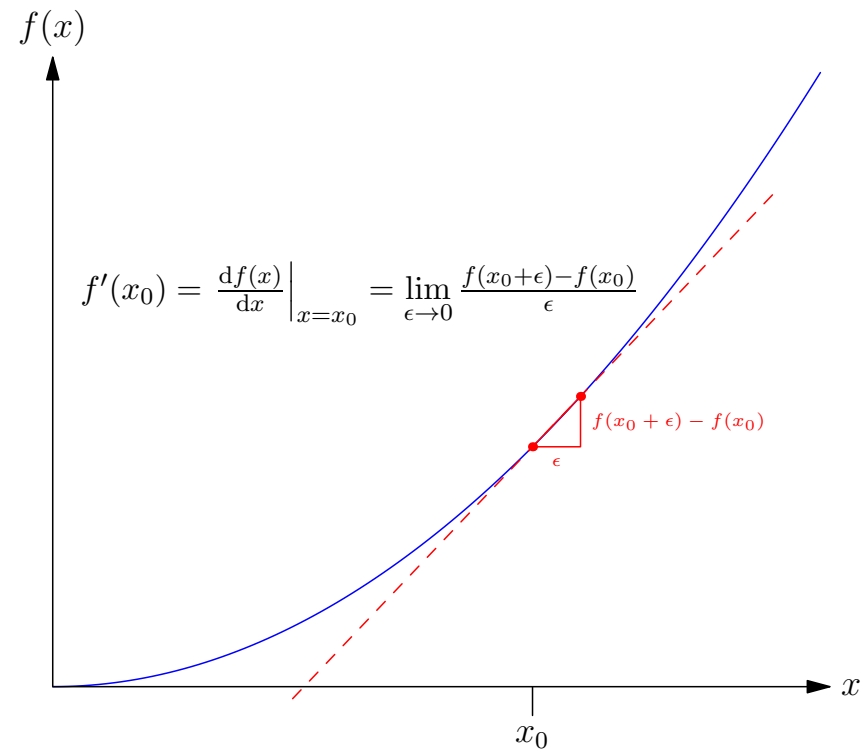
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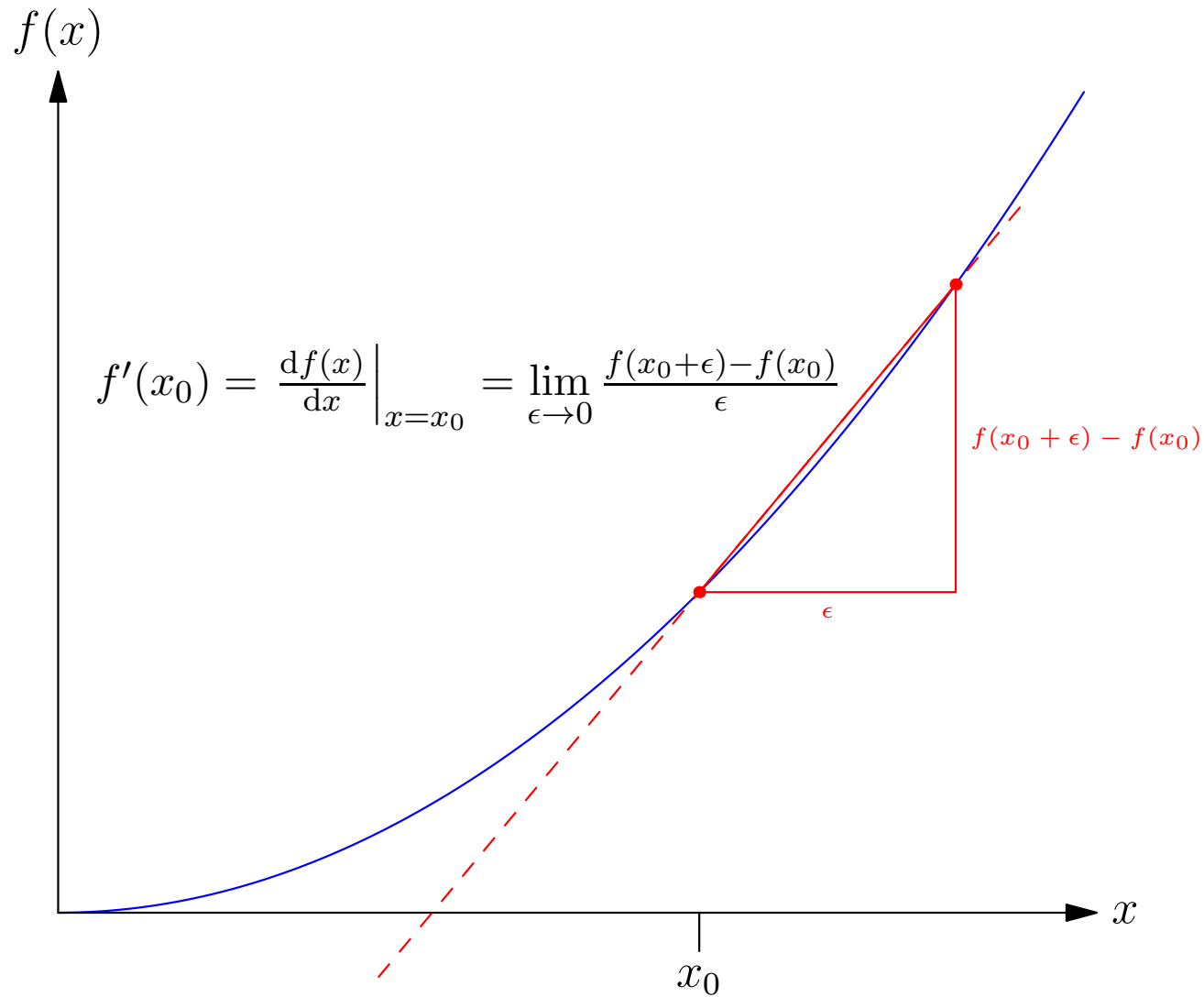
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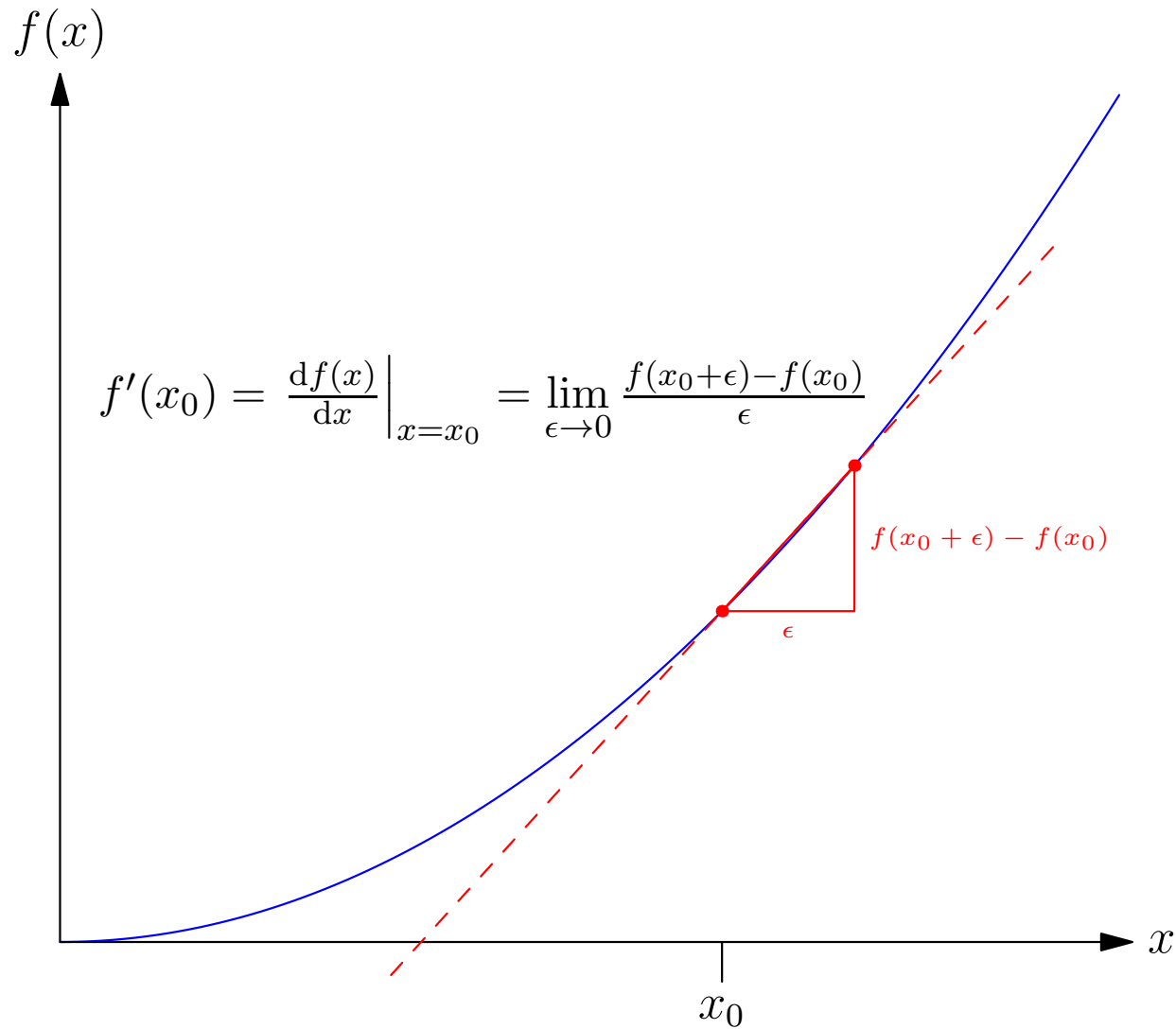
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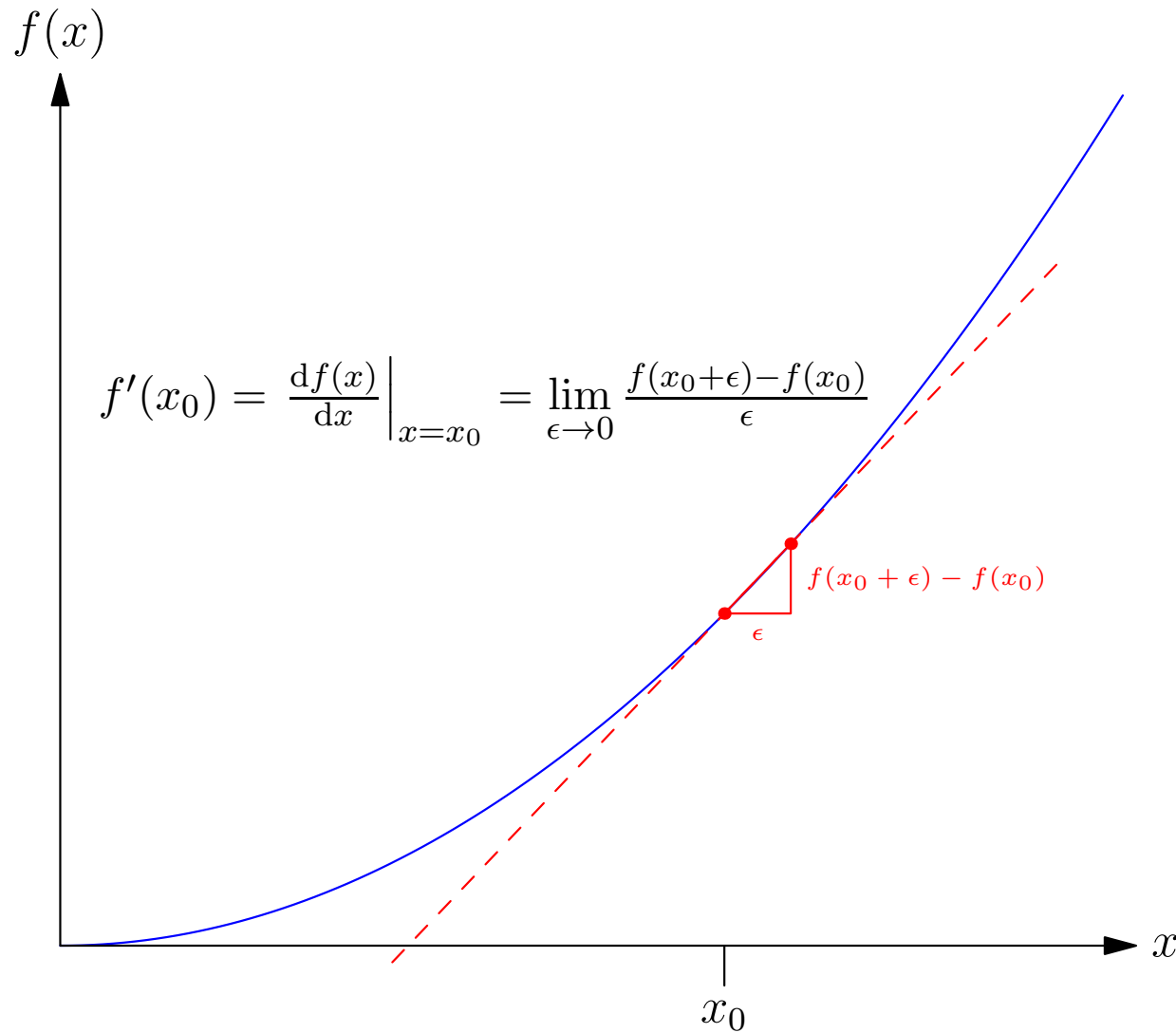
Differentiation



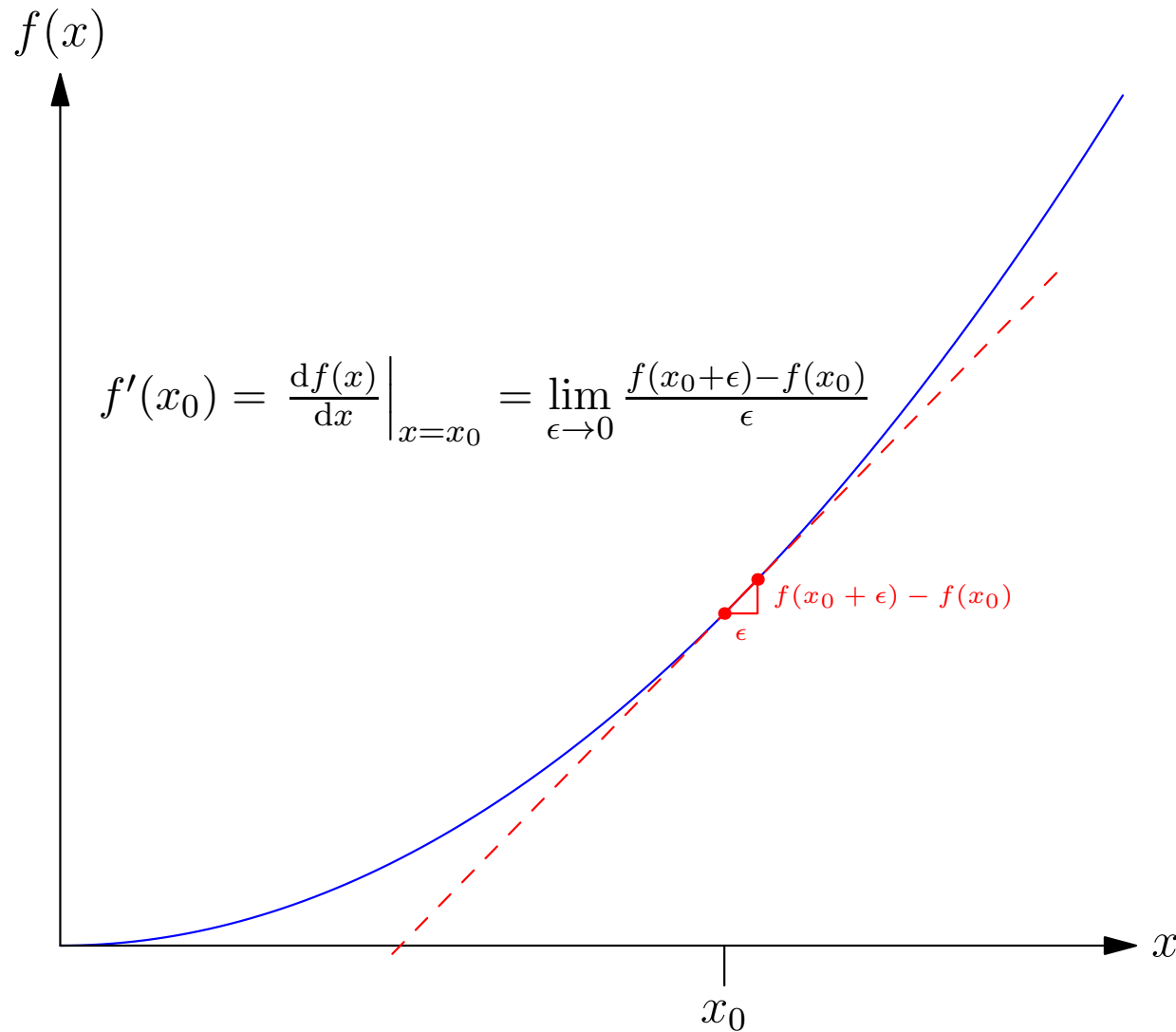
Differentiation



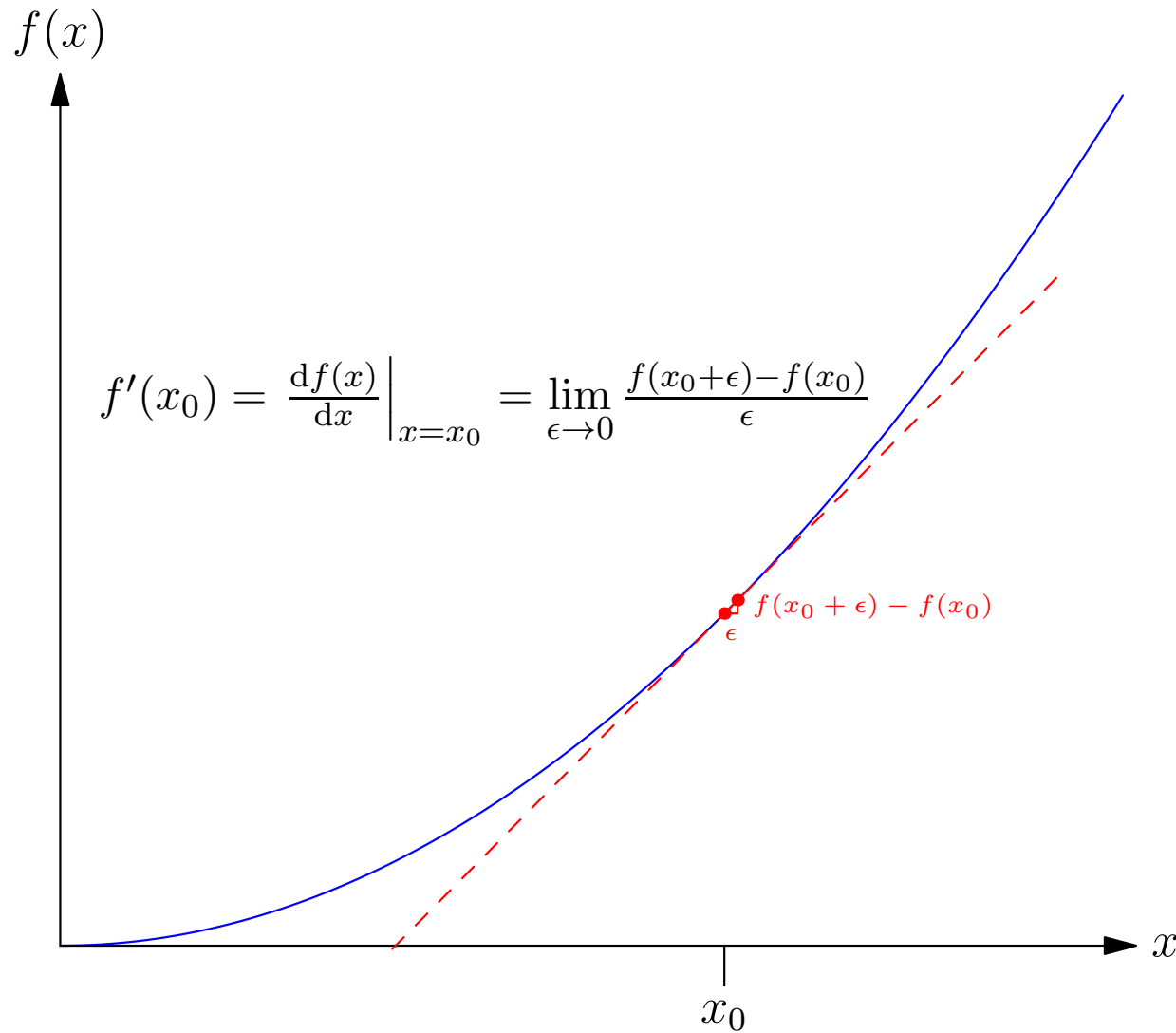
Differentiation



Differentiation



Differentiation



Polynomials

- $f(x) = x^2$

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Linearity of derivatives

- Consider the derivative of $af(x) + bg(x)$

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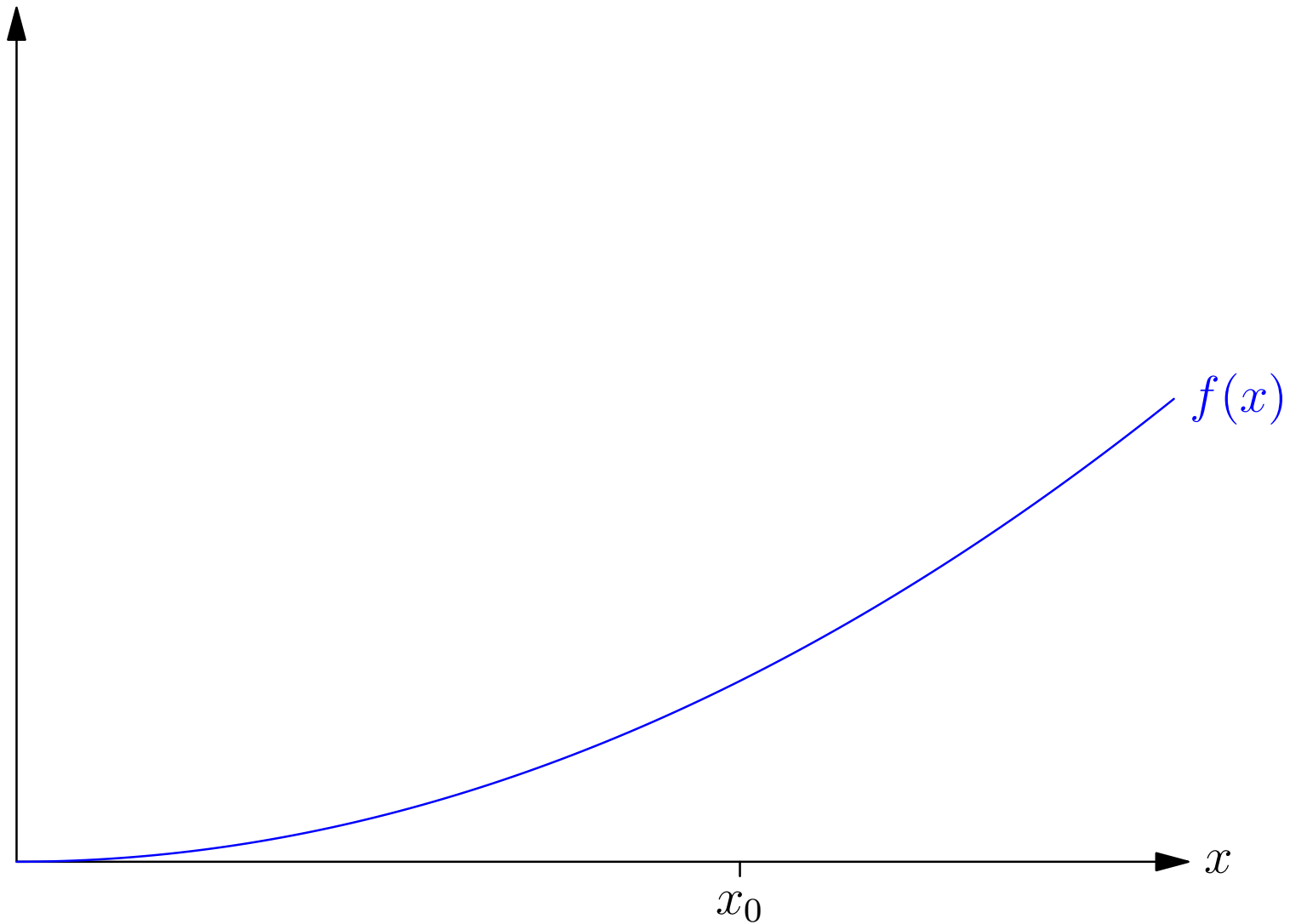
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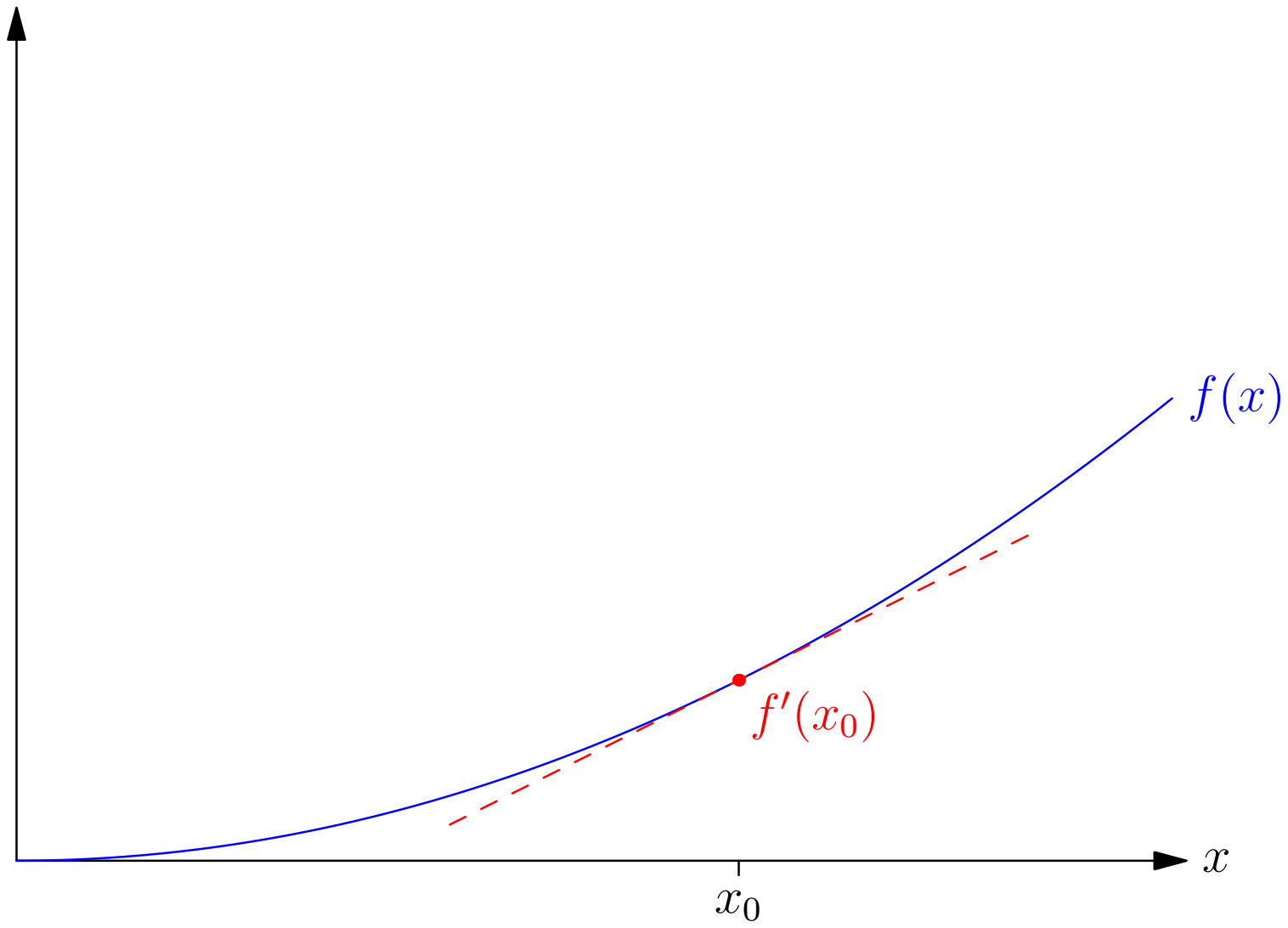
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- Differentiation is a linear operation!**

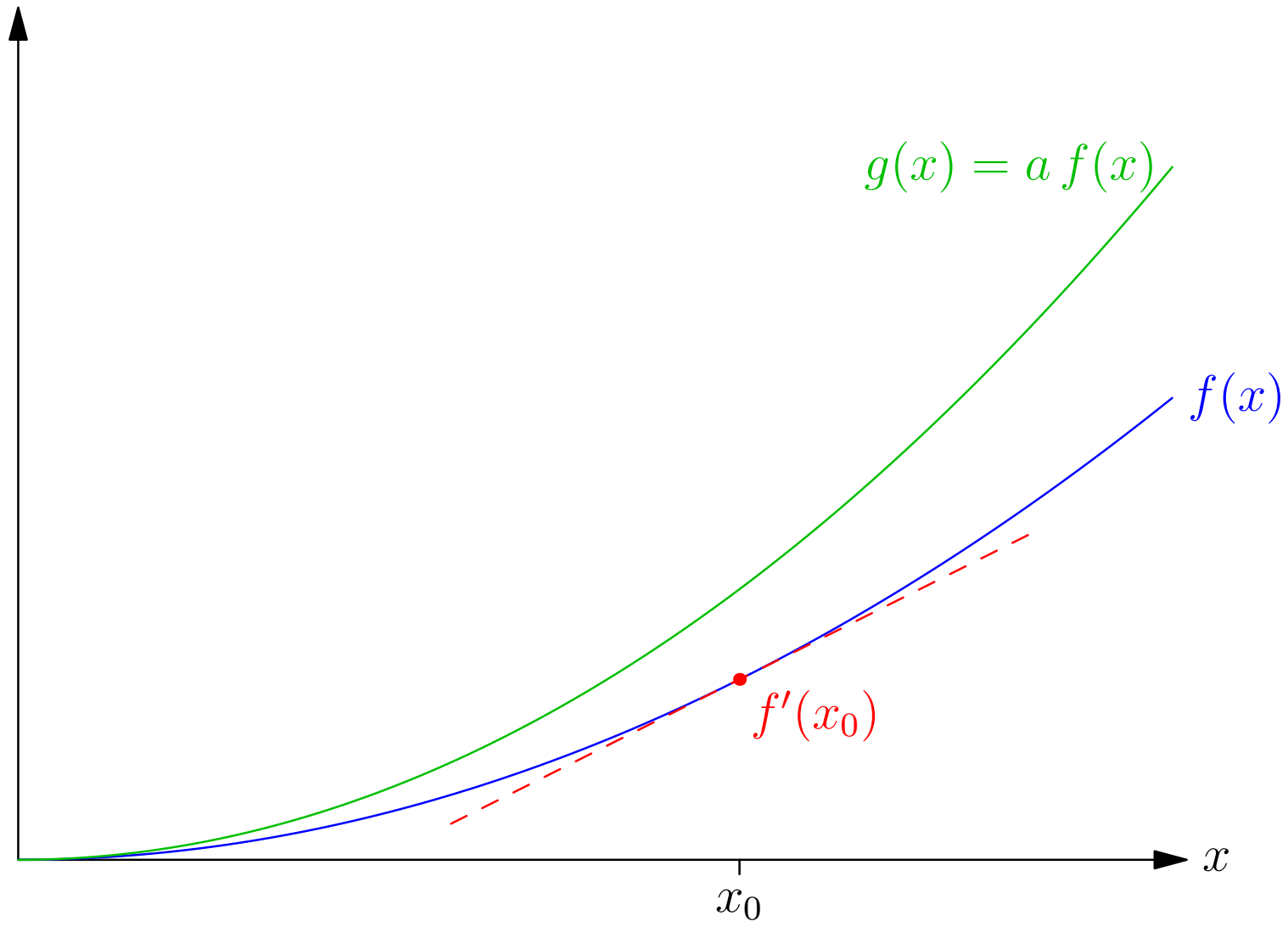
Linearity in Pictures



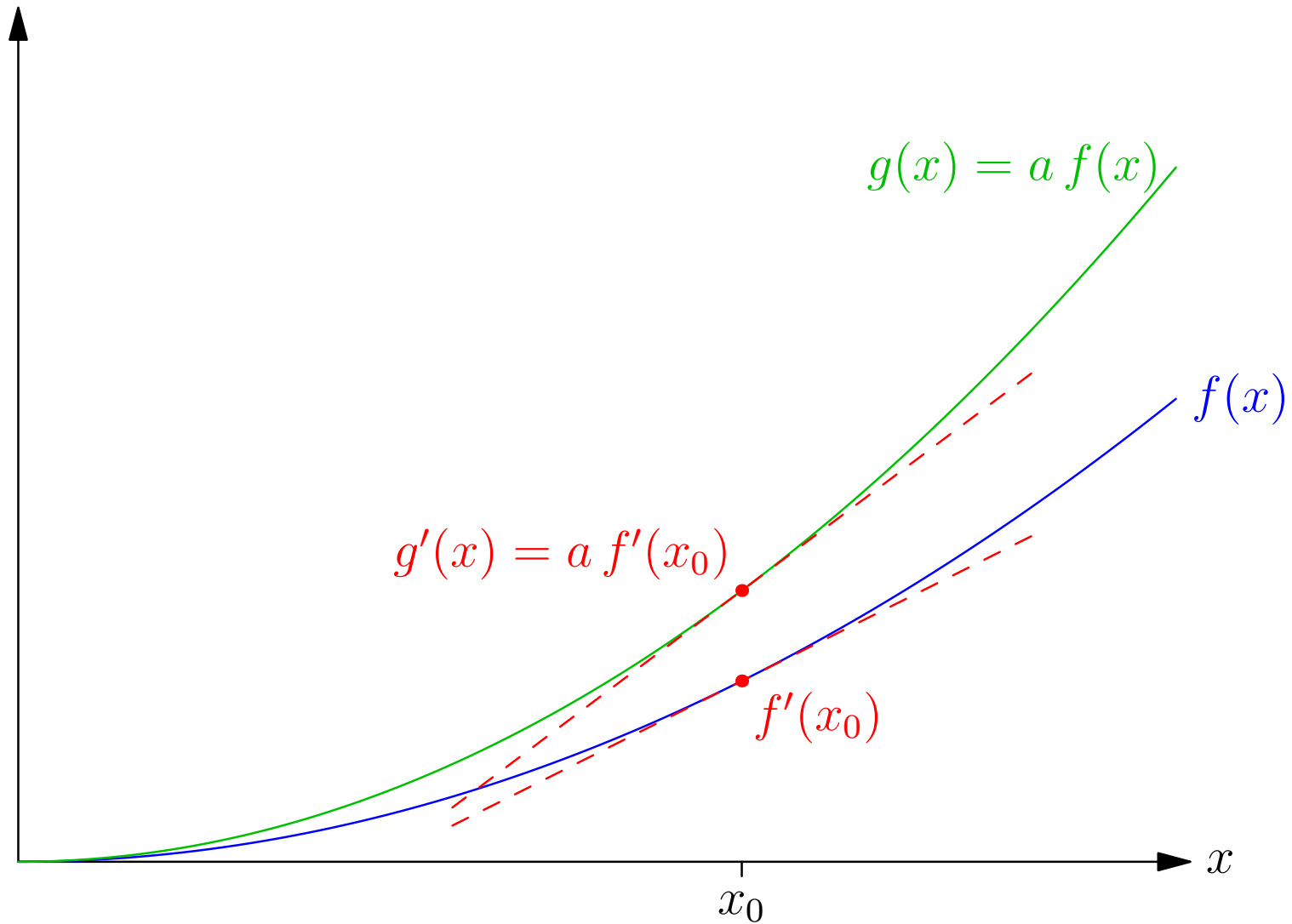
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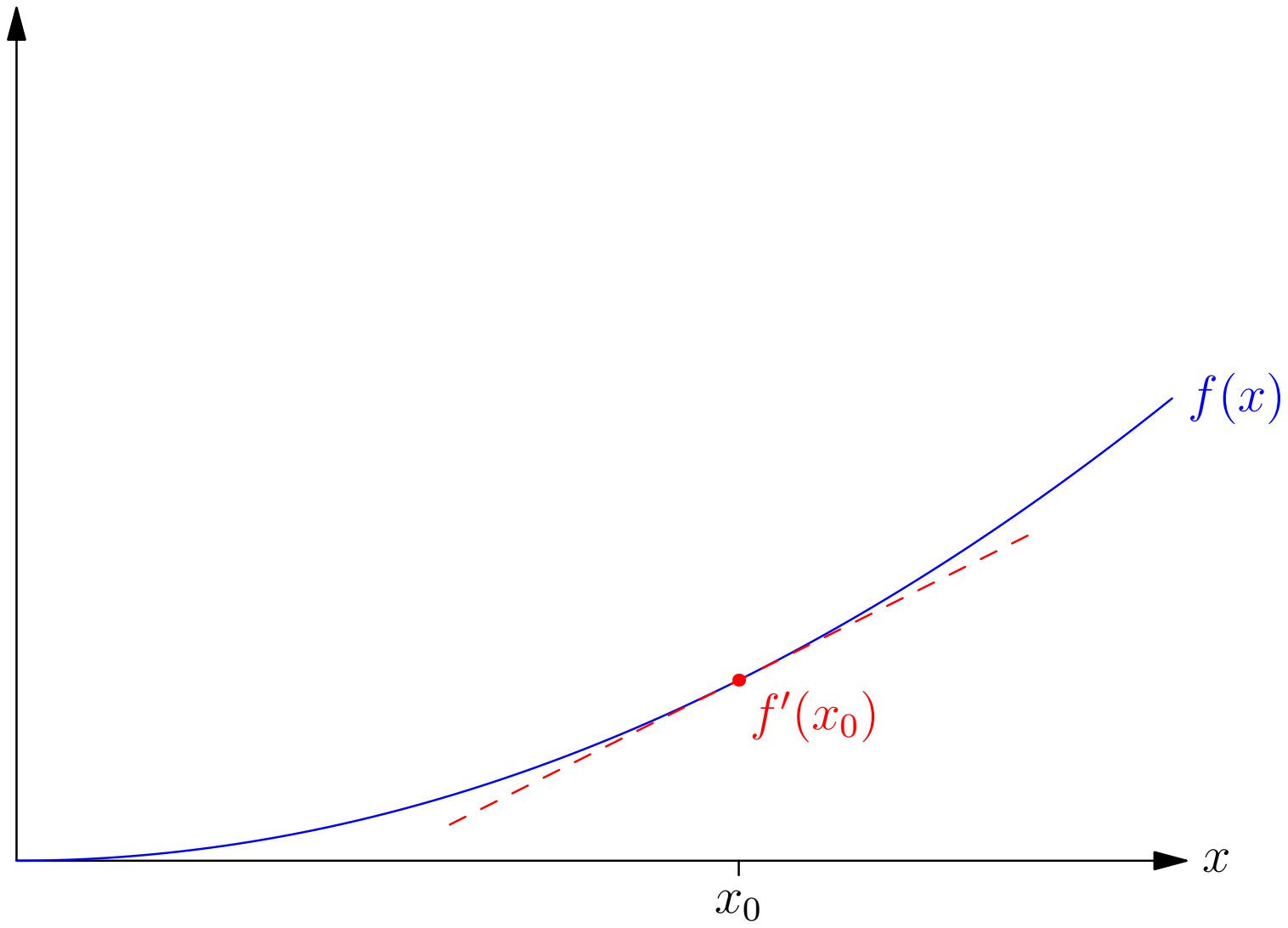
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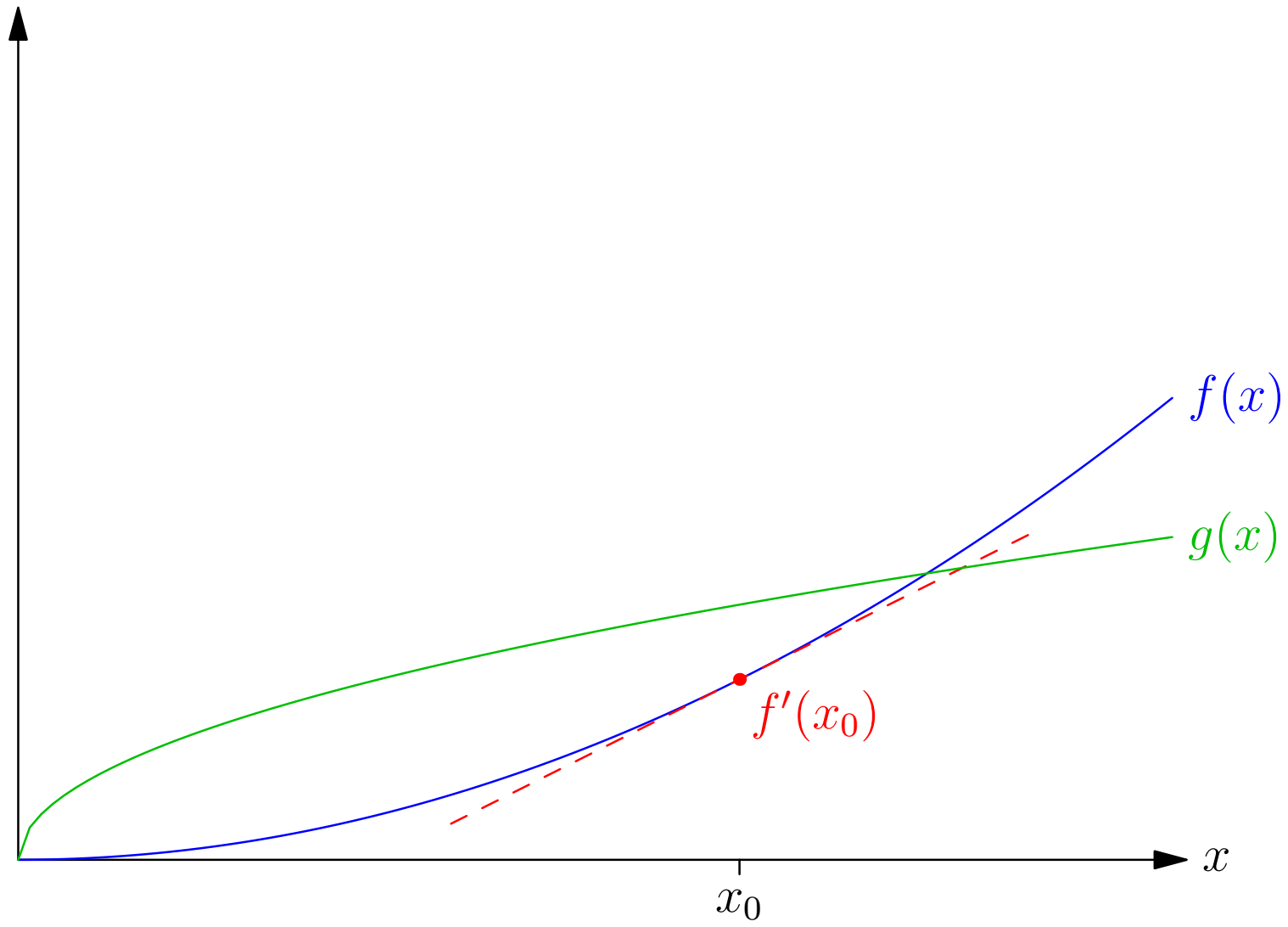
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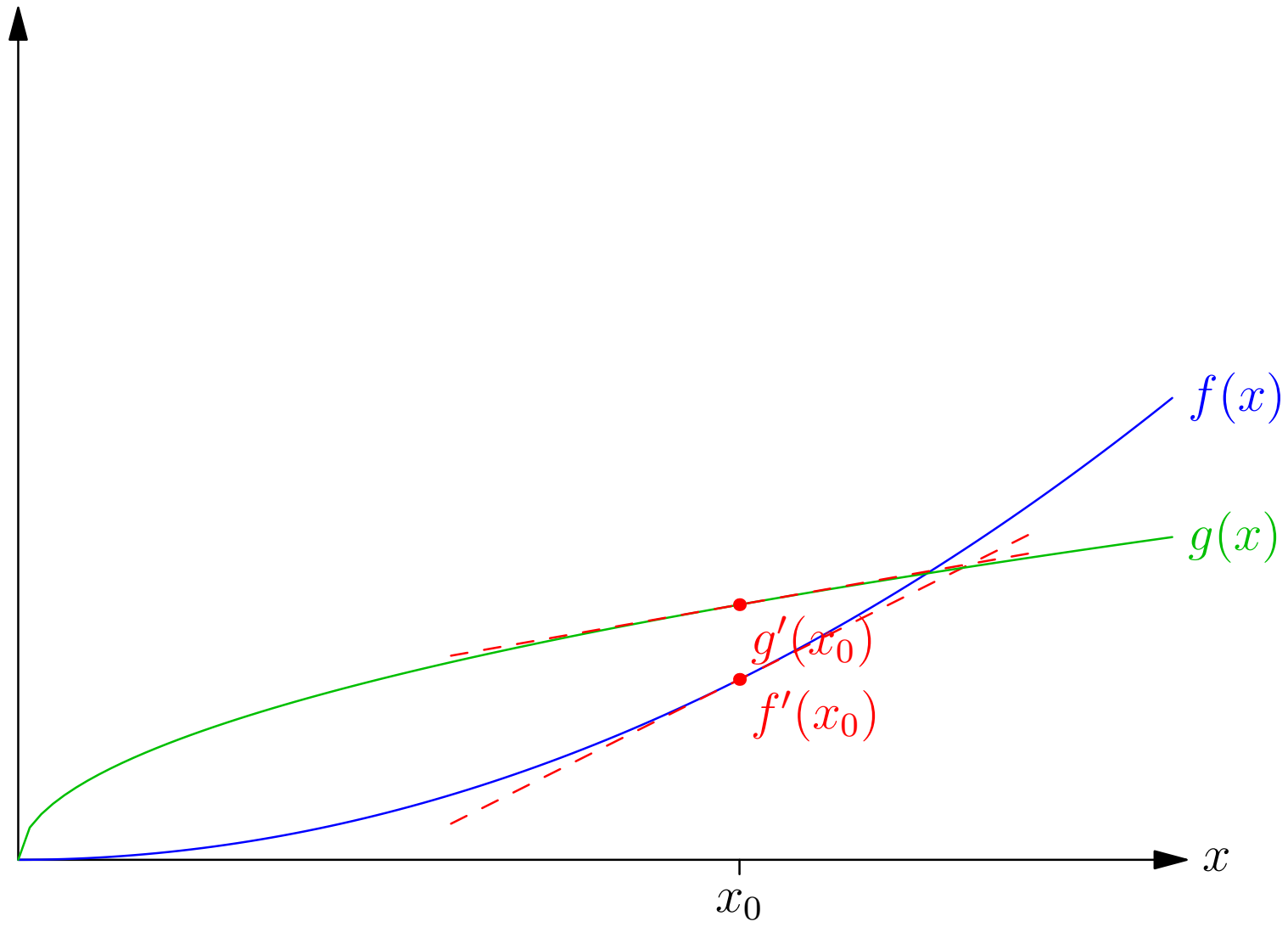
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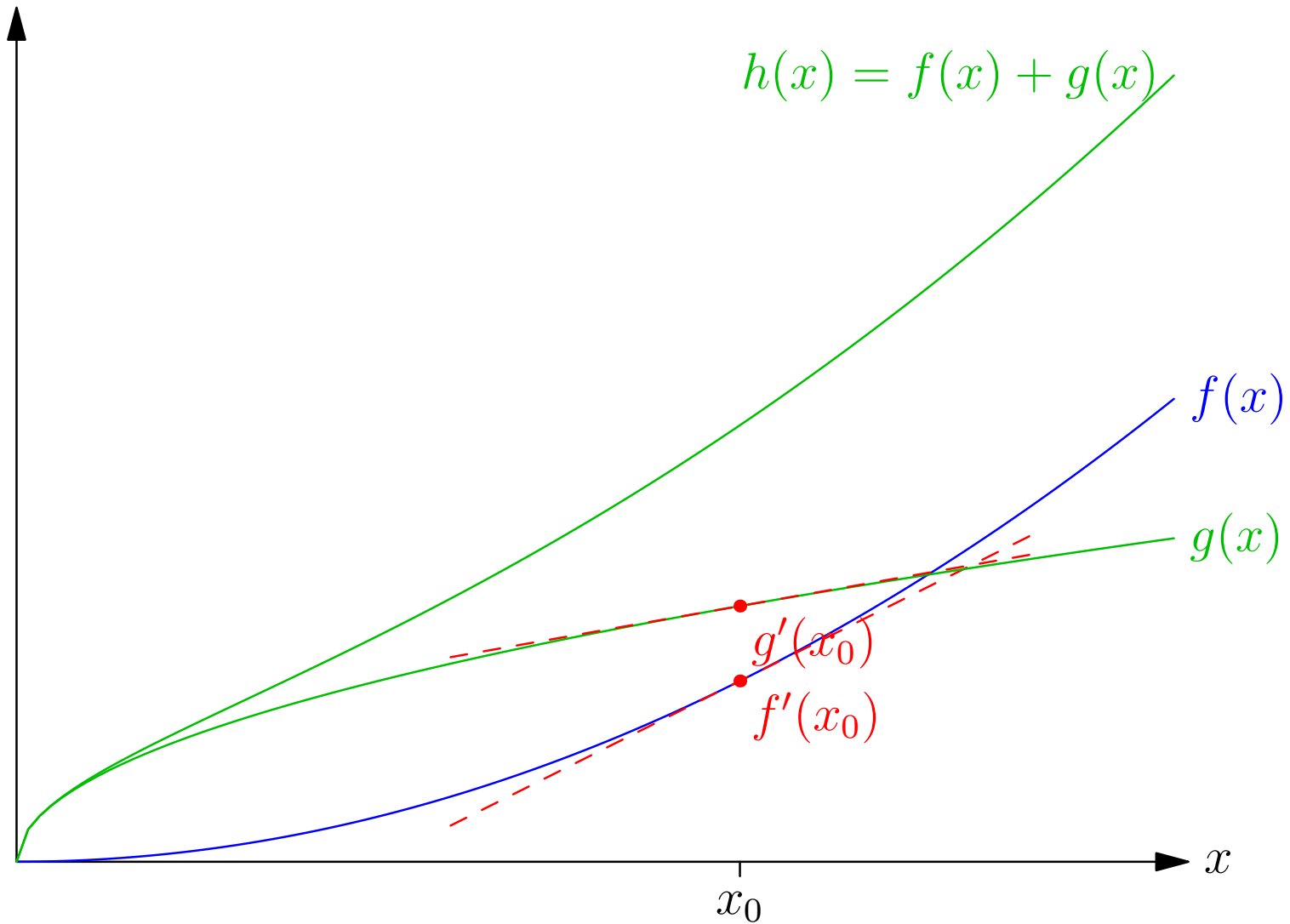
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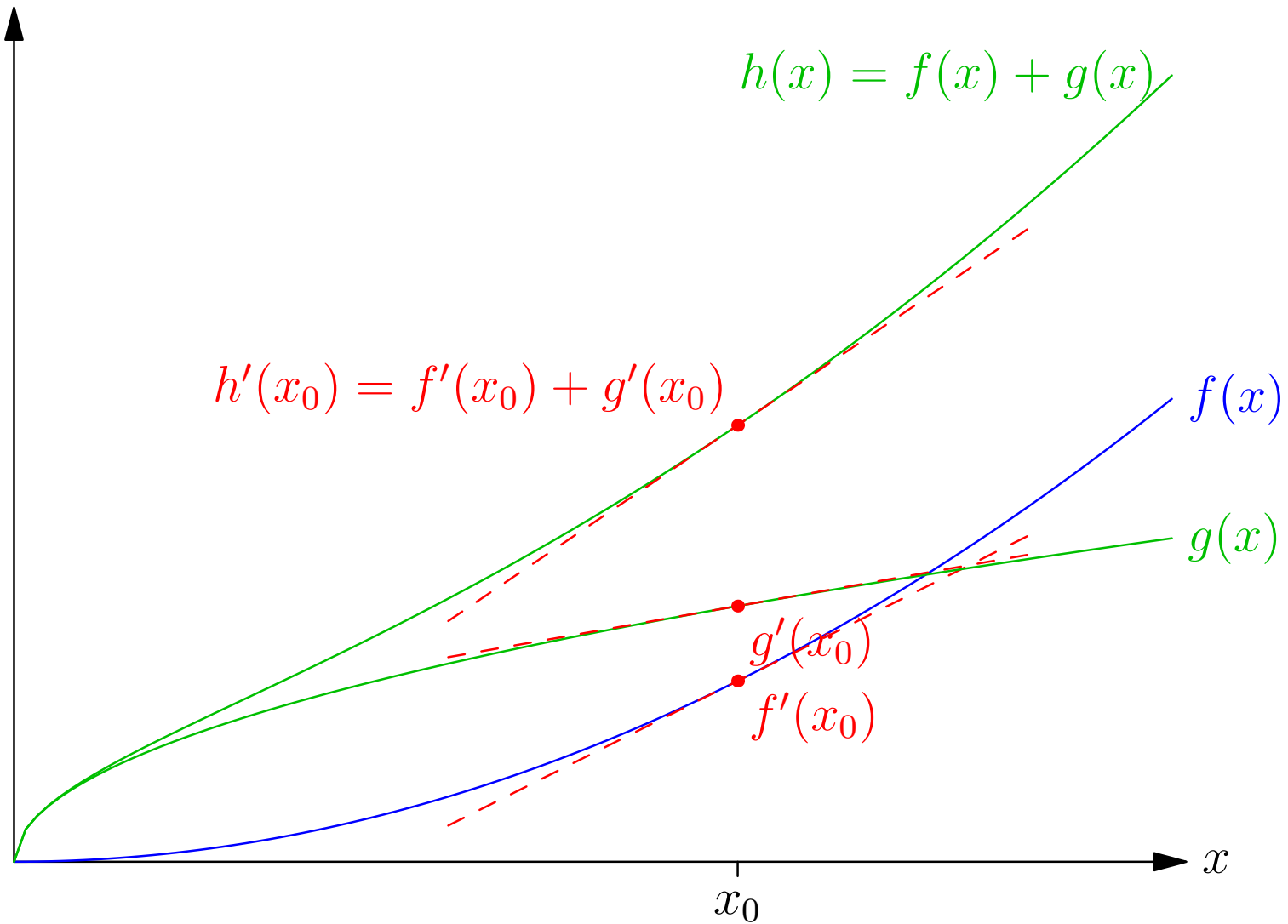
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- This is the **product rule**

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- This is the famous **chain rule**. Together with the product rule it means you can differentiate almost everything

More on chain rules

- We can also write the chain rule as

$$\frac{df(g(x))}{dx} = \frac{df(g)}{dg} \frac{dg(x)}{dx}$$

- Sometimes this is neater or easier to remember

$$\frac{de^{\cos(x^2)}}{dx} = \frac{de^{\cos(x^2)}}{d\cos(x^2)} \frac{d\cos(x^2)}{dx^2} \frac{dx^2}{dx}$$

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Inverse functions

- Suppose $g(y) = f^{-1}(y)$ is the inverse of $f(x)$ in the sense that $g(f(x)) = f^{-1}(f(x)) = x$
- Using the chain rule

$$\frac{dg(f(x))}{dx} = f'(x)g'(f(x))$$

- So $g'(f(x)) = 1/f'(x)$
- Writing $y = f(x)$ so that $x = f^{-1}(y) = g(y)$ we find $g'(y) = 1/f'(g(y))$ that is

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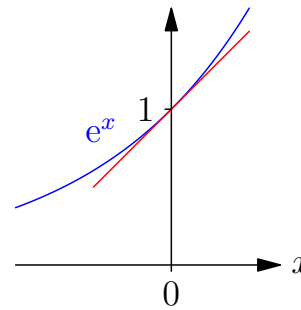
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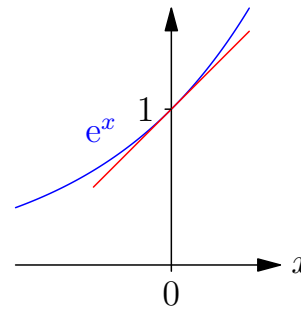
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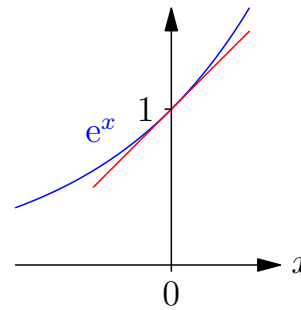


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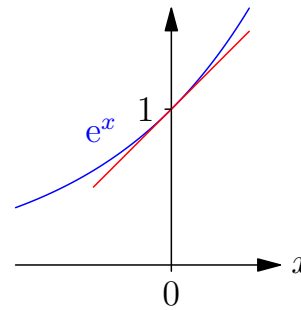


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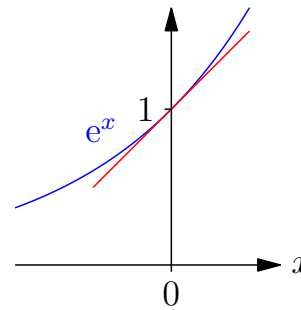
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Functions of Exponentials

- What about $f(x) = e^{cx}$

$$\frac{de^{cx}}{dx} = \frac{de^{cx}}{dcx} \frac{dcx}{dx}$$

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Natural Logarithms

- The natural logarithm is defined as the inverse of e^x

$$\ln(e^x) = x \qquad e^{\ln(y)} = y$$

- Recall that if $g(y) = f^{-1}(y)$ then $g'(y) = 1/f'(g(y))$
- Consider $g(y) = \ln(y)$ and $f(x) = e^x$ (with $f'(x) = e^x$)

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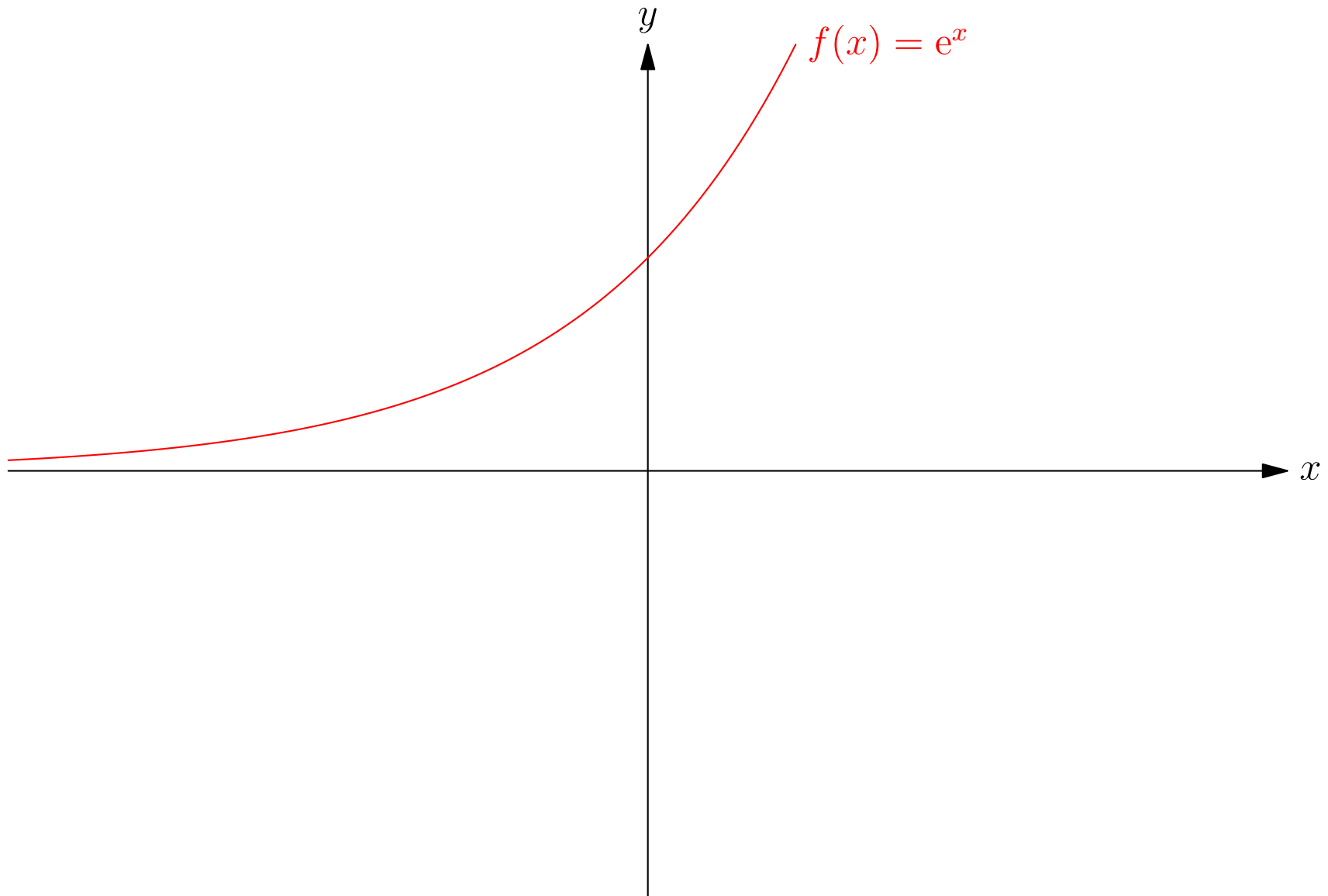
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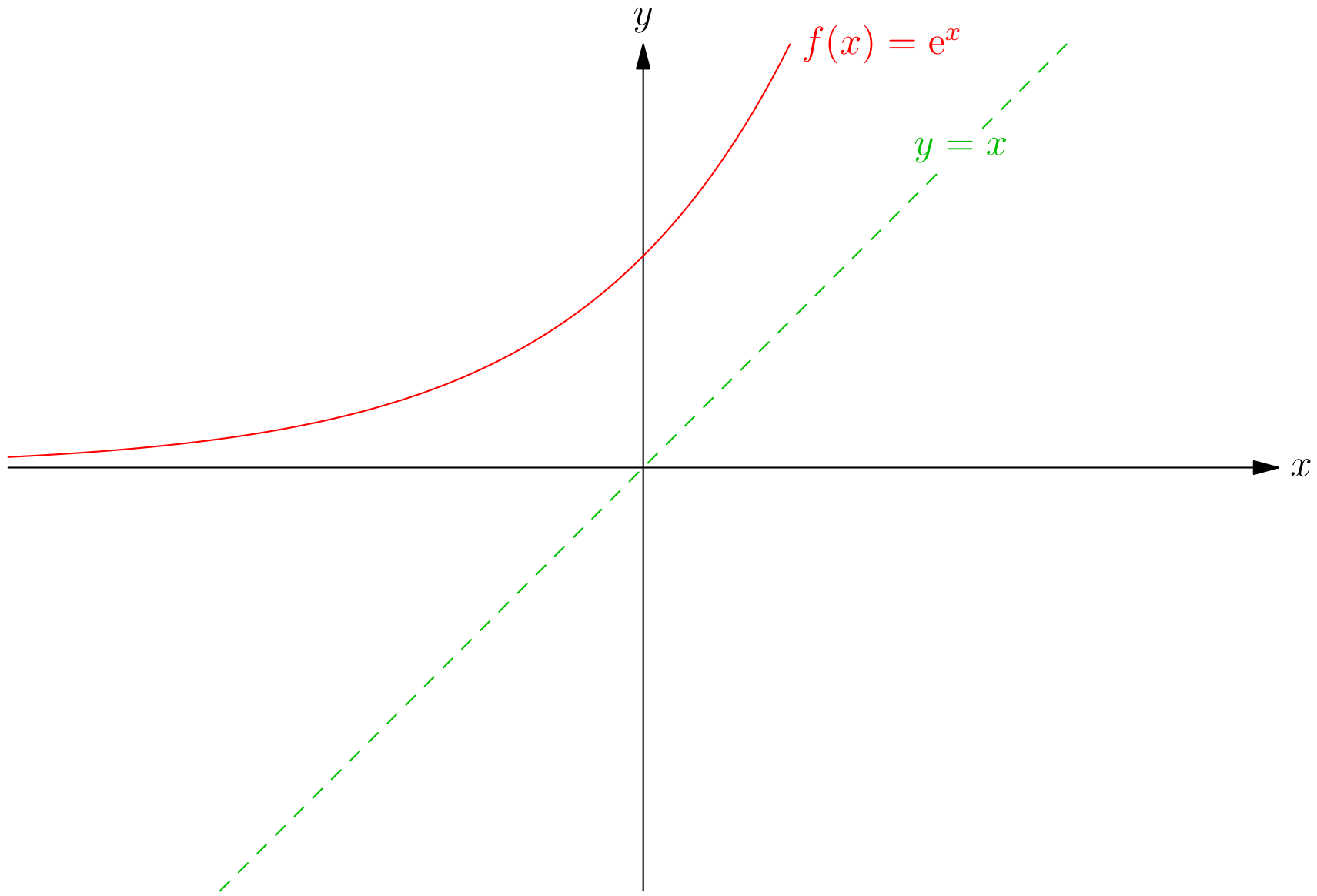
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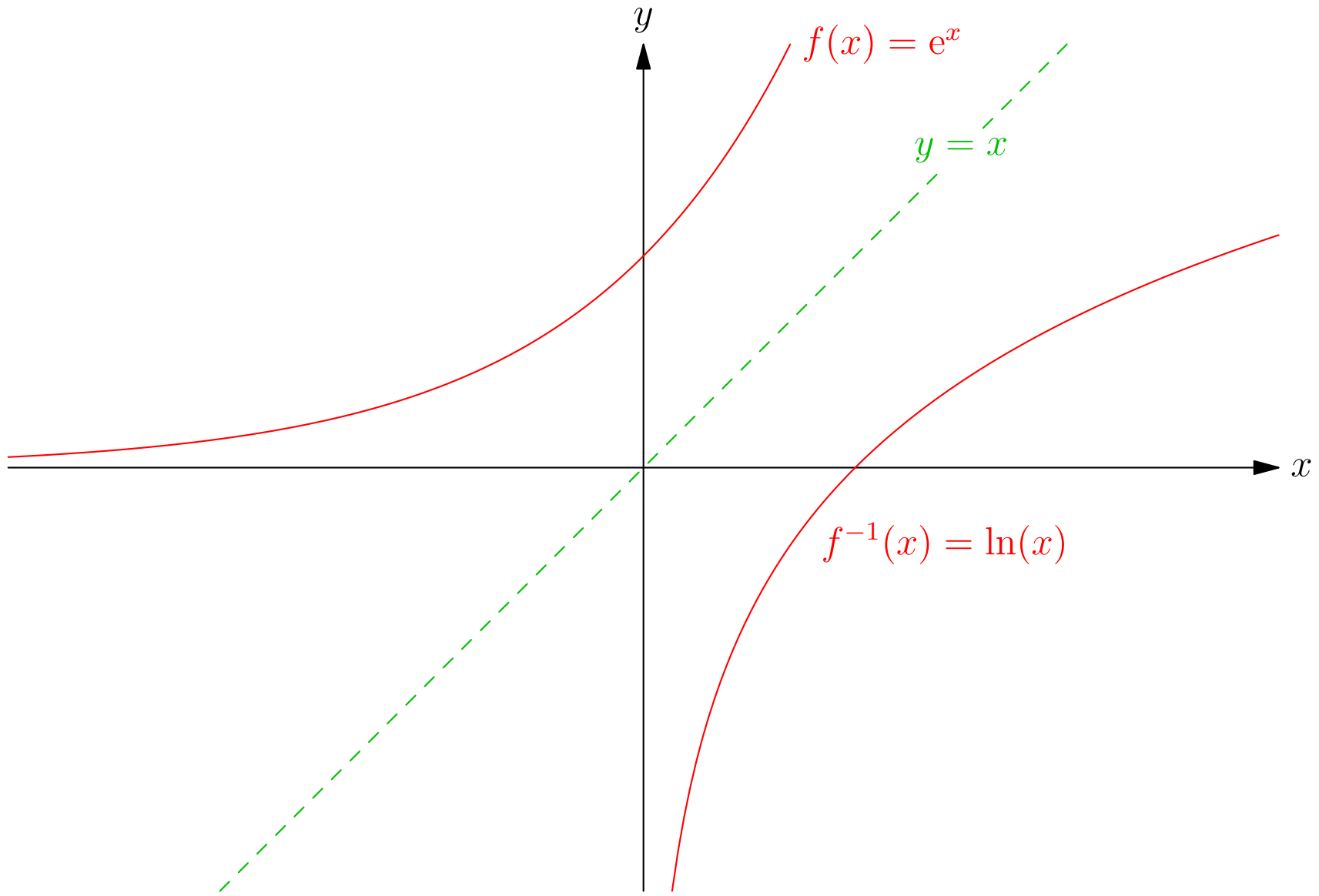
Exponentials and Logarithms



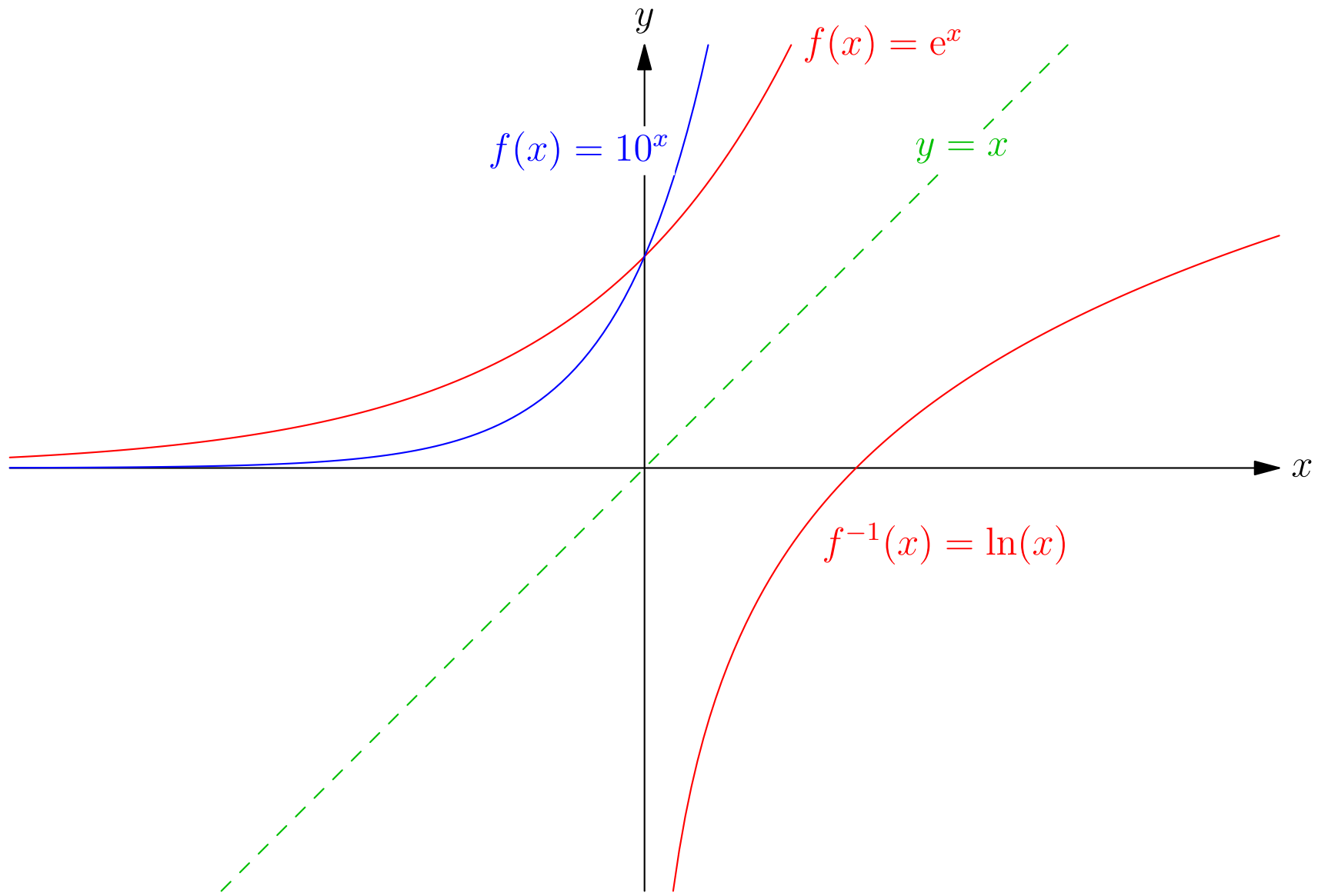
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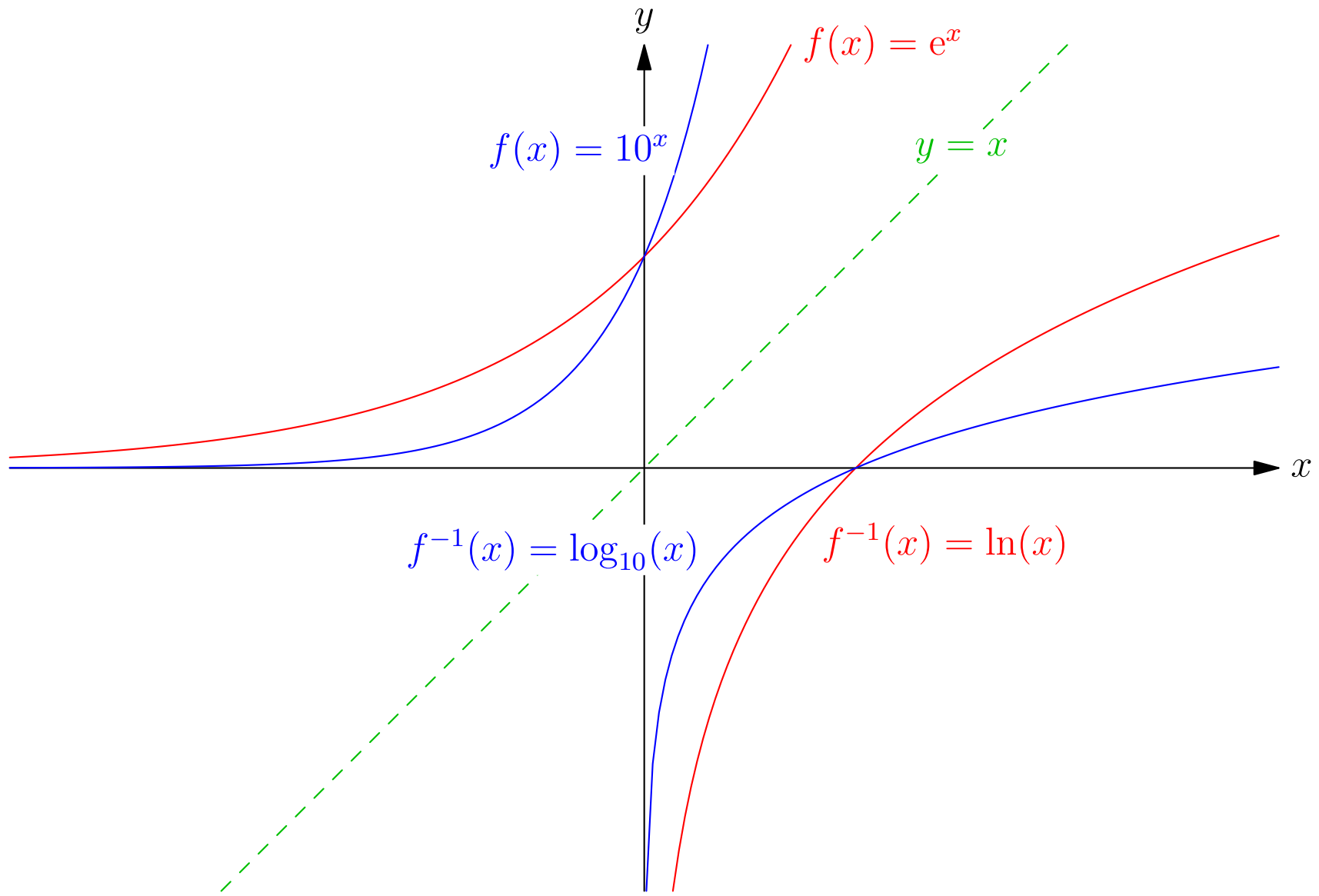
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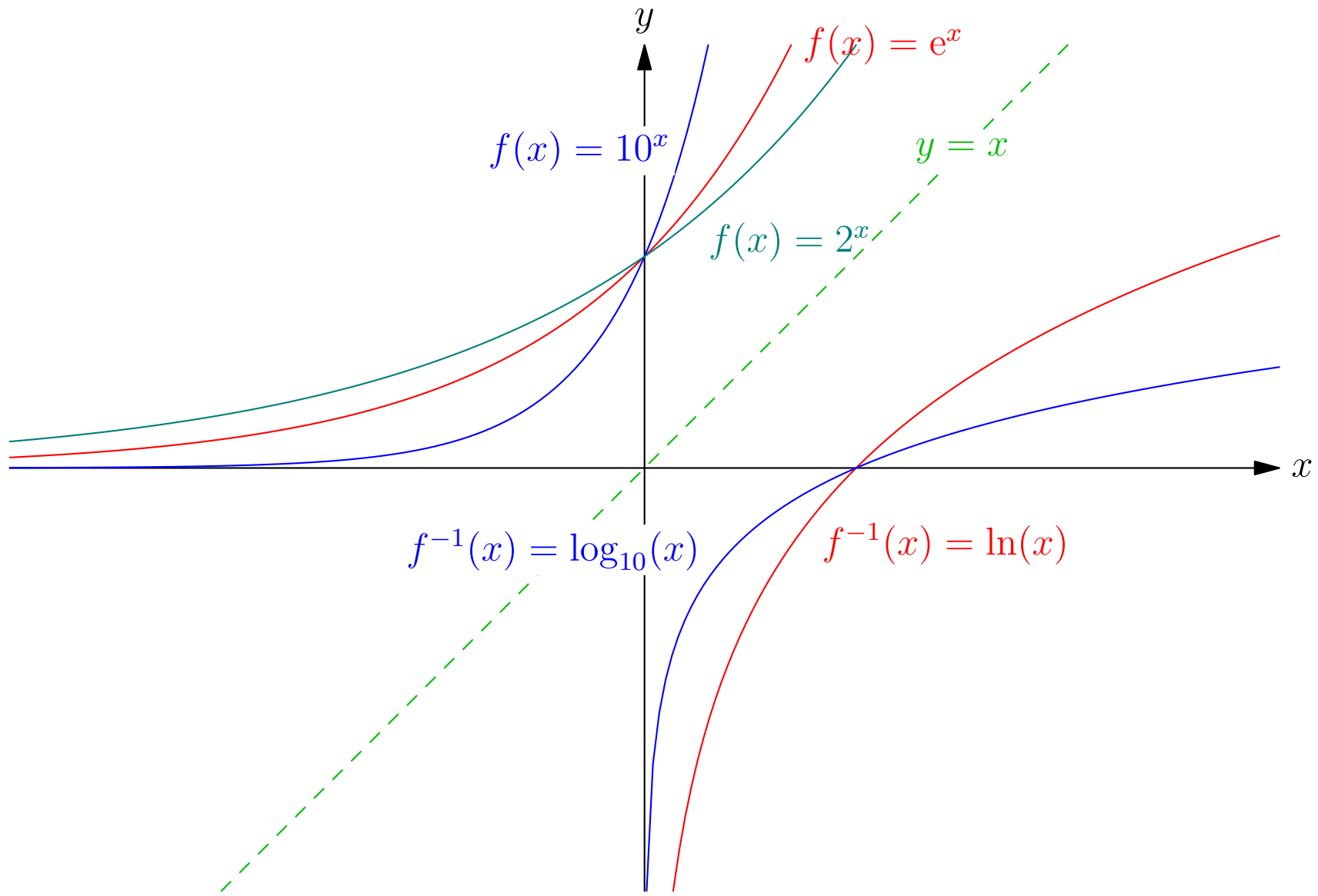
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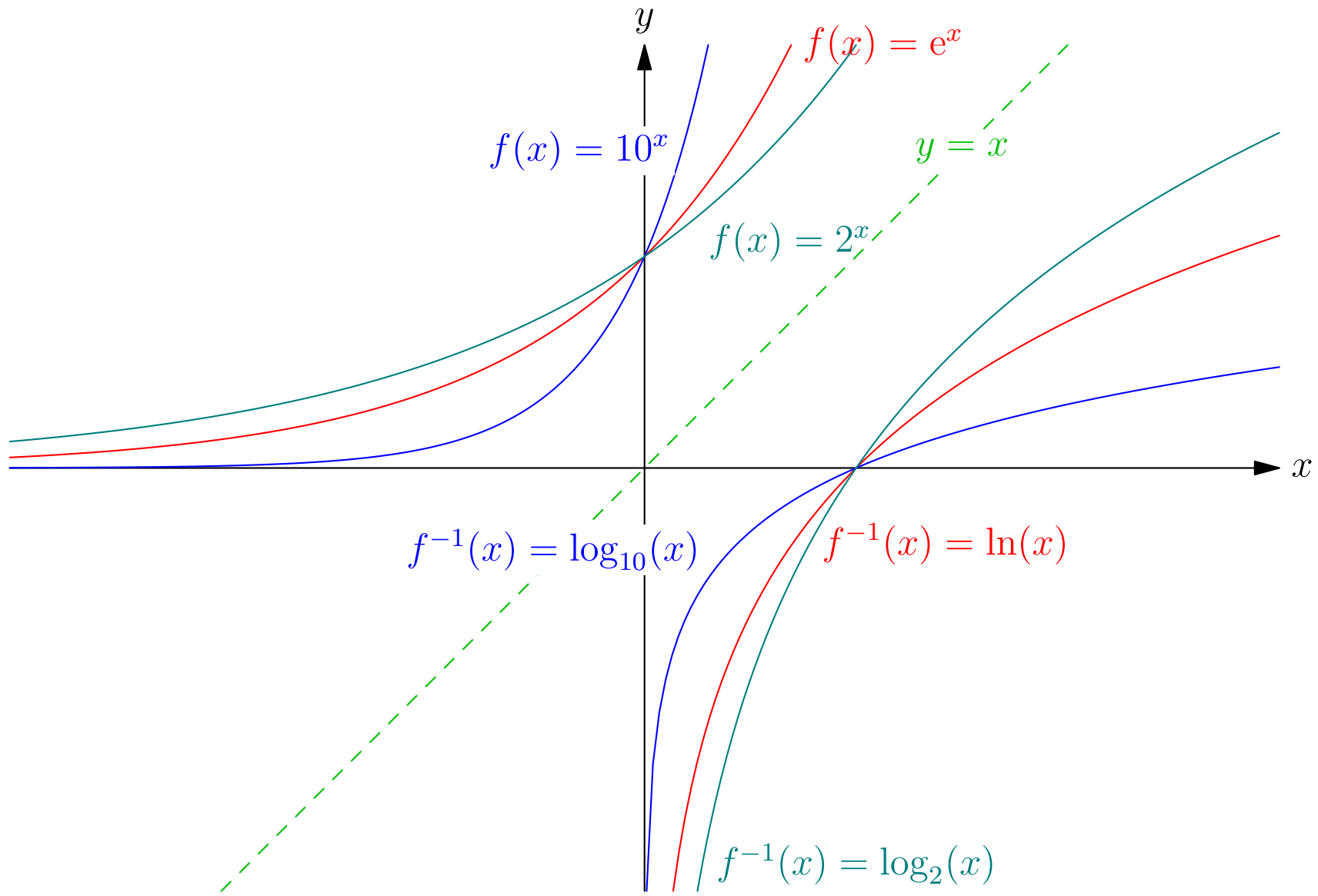
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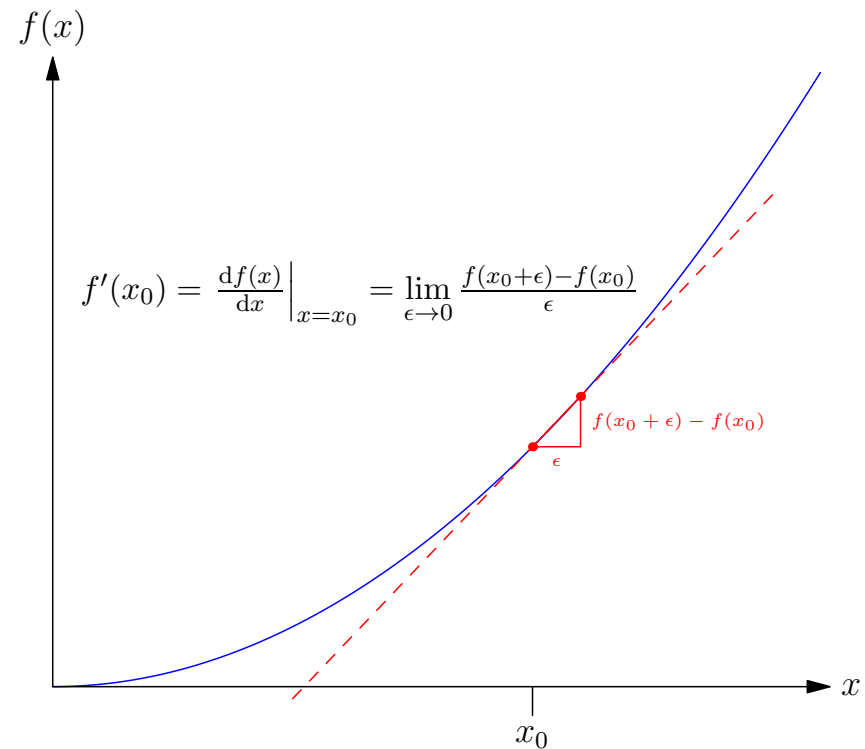


Exponentials and Logarithms



Outline

1. Why Calculus?
2. Differentiation
3. **Vector Calculus**
4. Matrix Calculus



Derivatives in High Dimensions

- When working with functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$ in many dimensions then there will typically be different derivative in different directions
- To compute the derivative in a direction $\mathbf{u} \in \mathbb{R}^n$ (where $\|\mathbf{u}\| = 1$) at a point $\mathbf{x} \in \mathbb{R}^n$ we use

$$\partial_{\mathbf{u}} F(\mathbf{x}) = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{x} + \epsilon \mathbf{u}) - f(\mathbf{x})}{\epsilon}$$

- If $\mathbf{u} = \boldsymbol{\delta}_i = (0, \dots, 0, 1, 0, \dots, 0)^\top$ (i.e. $u_i = 1$) then

$$\frac{\partial f(\mathbf{x})}{\partial x_i} = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{x} + \epsilon \boldsymbol{\delta}_i) - f(\mathbf{x})}{\epsilon}$$

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Taylor

- If we expand $f(\mathbf{x} + \epsilon \mathbf{u})$ to first order in ϵ

$$f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^\top \mathbf{g}(\mathbf{x}) + O(\epsilon^2)$$

then $g_i(\mathbf{x}) = \frac{\partial f(\mathbf{x})}{\partial x_i}$

- Recall we defined the vector of first order derivatives of $f(\mathbf{x})$ to be the gradient

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This is the start of the high-dimensional Taylor expansion

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- It is tedious to compute these things component-wise, but when you need to understand what is going on then go back to the basics

Computing Gradients 2

- A slicker way is just to expand $f(\mathbf{x} + \epsilon \mathbf{u})$
- Consider $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{M} \mathbf{x} + \mathbf{a}^\top \mathbf{x}$

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$$\nabla f(\mathbf{x}) = \mathbf{M} \mathbf{x} + \mathbf{M}^\top \mathbf{x} + \mathbf{a}$$

Function Composition

- The gradient tells us about the change in a function value when we perturb an input $\mathbf{x}$ by a small amount in the directions $\mathbf{u}$

$$f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^\top \nabla f(\mathbf{x}) + O(\epsilon^2)$$

- What happens when we compose functions: $h(\mathbf{x}) = f(\mathbf{g}(\mathbf{x}))$.
Note that,

$$\mathbf{x} \in \mathbb{R}^n \quad \mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^m \quad f : \mathbb{R}^m \rightarrow \mathbb{R} \quad h : \mathbb{R}^n \rightarrow \mathbb{R}$$

- The vector valued function $\mathbf{g}(\mathbf{x})$ can be seen as m independent

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$$\mathbf{x} \in \mathbb{R}^n \quad \mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^m \quad f : \mathbb{R}^m \rightarrow \mathbb{R} \quad h : \mathbb{R}^n \rightarrow \mathbb{R}$$

- The vector valued function $\mathbf{g}(\mathbf{x})$ can be seen as m independent

$$\text{function } \mathbf{g}(\mathbf{x}) = \begin{pmatrix} g_1(\mathbf{x}) \\ g_2(\mathbf{x}) \\ \vdots \\ g_m(\mathbf{x}) \end{pmatrix}$$

Function Composition

- The gradient tells us about the change in a function value when we perturb an input $\mathbf{x}$ by a small amount in the directions $\mathbf{u}$

$$f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^\top \nabla f(\mathbf{x}) + O(\epsilon^2)$$

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Jacobians

- Perturbing $\mathbf{x}$ by $\epsilon \mathbf{u}$

$$g_i(\mathbf{x} + \epsilon \mathbf{u}) = g_i(\mathbf{x}) + \epsilon \mathbf{u}^\top \nabla g_i(\mathbf{x}) + O(\epsilon^2)$$

so that

$$\mathbf{g}(\mathbf{x} + \epsilon \mathbf{u}) = \mathbf{g}(\mathbf{x}) + \epsilon \mathbf{J} \mathbf{u} + O(\epsilon^2)$$

where $\mathbf{J}$ is an $m \times n$ matrix with components $J_{ij} = \frac{\partial g_i(\mathbf{x})}{\partial x_j}$

- Note that $\mathbf{J} \mathbf{u}$ is a column vector of size m
- Ignoring terms of order ϵ^2

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Chain Rule in High Dimensions

- Using $h(\mathbf{x}) = f(\mathbf{g}(\mathbf{x}))$

$$h(\mathbf{x} + \epsilon \mathbf{u}) = h(\mathbf{x}) + \epsilon \mathbf{u}^\top \mathbf{J}^\top \nabla_{\mathbf{g}} f(\mathbf{g}(\mathbf{x}))$$

- We get that

$$\nabla_{\mathbf{x}} h(\mathbf{x}) = \nabla_{\mathbf{x}} f(\mathbf{g}(\mathbf{x})) = \mathbf{J}^\top \nabla_{\mathbf{g}} f(\mathbf{g}(\mathbf{x}))$$

- Componentwise

$$\frac{\partial h(\mathbf{x})}{\partial x_i} = \frac{\partial f(\mathbf{g}(\mathbf{x}))}{\partial x_i} = \sum_{j=1}^m \frac{\partial g_j(\mathbf{x})}{\partial x_i} \frac{\partial f(\mathbf{g}(\mathbf{x}))}{\partial g_j}$$

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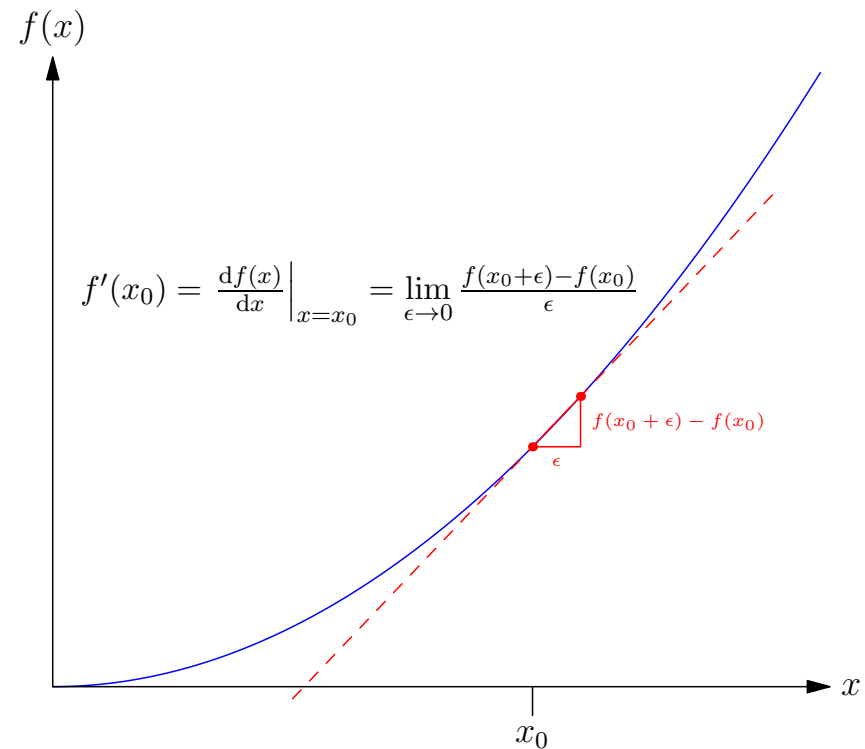
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Outline

1. Why Calculus?
2. Differentiation
3. Vector Calculus
4. **Matrix Calculus**



Differentiating Matrices

- Often we have loss functions with respect to a matrix $\mathbf{W}$, e.g.

$$L(\mathbf{W}) = (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c)^2$$

- We might want to find the minimum with respect to $\mathbf{W}$
- This occurs at a point $\mathbf{W}^*$ where $L(\mathbf{W})$ does not increase as we change $\mathbf{W}$ in any way
- That is, we seek a $\mathbf{W}^*$ such that, for any matrices $\mathbf{U}$

$$L(\mathbf{W}^* + \epsilon \mathbf{U}) - L(\mathbf{W}^*) = O(\epsilon^2)$$

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Generalised Gradient

- We can generalise the idea of gradient to matrices

$$\frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} = \begin{pmatrix} \frac{\partial L(\mathbf{W})}{\partial W_{11}} & \frac{\partial L(\mathbf{W})}{\partial W_{12}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{1m}} \\ \frac{\partial L(\mathbf{W})}{\partial W_{21}} & \frac{\partial L(\mathbf{W})}{\partial W_{22}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{2m}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial L(\mathbf{W})}{\partial W_{n1}} & \frac{\partial L(\mathbf{W})}{\partial W_{n2}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{nm}} \end{pmatrix}$$

- From an identical argument we used for vectors

$$L(\mathbf{W} + \epsilon \mathbf{U}) = L(\mathbf{W}) + \epsilon \text{tr} \mathbf{U}^\top \frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} + O(\epsilon^2)$$

where

$$\text{tr} \mathbf{U}^\top \mathbf{G} = \sum_i [\mathbf{U}^\top \mathbf{G}]_{ii} = \sum_{ij} U_{ji} G_{ji} = \sum_{ij} U_{ij} G_{ij} = \langle \mathbf{U}, \mathbf{G} \rangle$$

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Example

- Suppose

$$L(\mathbf{W}) = (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c)^2$$

then

$$L(\mathbf{W} + \epsilon \mathbf{U}) = (\mathbf{a}^\top (\mathbf{W} + \epsilon \mathbf{U}) \mathbf{b} - c)^2$$

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$$\mathbf{a}^\top \mathbf{U} \mathbf{b} = \sum_{ij} a_i U_{ij} b_j = \sum_{ij} U_{ji} a_j b_i = \text{tr} \mathbf{U}^\top \mathbf{a} \mathbf{b}^\top$$

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$$\text{Thus } \frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} = 2 (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c) \mathbf{a} \mathbf{b}^\top$$

Traces

- The trace of a matrix is the sum of its diagonal elements

$$\text{tr} \mathbf{A} = \text{tr} \mathbf{A}^T = \sum_i A_{ii}$$

- Clearly $\text{tr} c\mathbf{A} = c \text{tr} \mathbf{A}$
- Also $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr} \mathbf{A} + \text{tr} \mathbf{B}$
- We note that

$$\text{tr} \mathbf{A} \mathbf{B} = \sum_{i,j} A_{ij} B_{ji}$$

- It follows that

$$\text{tr} \mathbf{A} \mathbf{B} \mathbf{C} \mathbf{D} = \text{tr} \mathbf{D} \mathbf{A} \mathbf{B} \mathbf{C}$$

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$$\text{tr}\mathbf{AB} = \sum_{i,j} A_{ij}B_{ji}$$

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$$\text{tr}\mathbf{ABCD} = \text{tr}\mathbf{DABC}$$

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- The trace of a matrix is the sum of its diagonal elements

$$\text{tr}\mathbf{A} = \text{tr}\mathbf{A}^\top = \sum_i A_{ii}$$

- Clearly $\text{tr}c\mathbf{A} = c\text{tr}\mathbf{A}$
- Also $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}\mathbf{A} + \text{tr}\mathbf{B}$
- We note that

$$\text{tr}\mathbf{A}\mathbf{B} = \sum_{i,j} A_{ij}B_{ji} = \sum_{i,j} B_{ij}A_{ji} = \text{tr}\mathbf{B}\mathbf{A}$$

- It follows that

$$\text{tr}\mathbf{A}\mathbf{B}\mathbf{C}\mathbf{D} = \text{tr}\mathbf{D}\mathbf{A}\mathbf{B}\mathbf{C} = \text{tr}\mathbf{C}\mathbf{D}\mathbf{A}\mathbf{B} = \text{tr}\mathbf{B}\mathbf{C}\mathbf{D}\mathbf{A}$$

Quick Matrix Differentiation

- Let

$$\partial_{\mathbf{U}} f(\mathbf{X}) = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{X} + \epsilon \mathbf{U}) - f(\mathbf{X})}{\epsilon}$$

- E.g.

$$\begin{aligned} \partial_{\mathbf{U}} \operatorname{tr} \mathbf{A} \mathbf{X} \mathbf{B} &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (\operatorname{tr} \mathbf{A} (\mathbf{X} + \epsilon \mathbf{U}) \mathbf{B} - \operatorname{tr} \mathbf{A} \mathbf{X} \mathbf{B}) \\ &= \operatorname{tr} \mathbf{A} \mathbf{U} \mathbf{B} \end{aligned}$$

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Quick Matrix Differentiation

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$$\partial_{\mathbf{U}} f(\mathbf{X}) = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{X} + \epsilon \mathbf{U}) - f(\mathbf{X})}{\epsilon} = \text{tr } \mathbf{U}^T \frac{\partial f(\mathbf{X})}{\partial \mathbf{X}}$$

- E.g.

$$\begin{aligned} \partial_{\mathbf{U}} \text{tr } \mathbf{A} \mathbf{X} \mathbf{B} &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (\text{tr } \mathbf{A} (\mathbf{X} + \epsilon \mathbf{U}) \mathbf{B} - \text{tr } \mathbf{A} \mathbf{X} \mathbf{B}) \\ &= \text{tr } \mathbf{A} \mathbf{U} \mathbf{B} = \text{tr } \mathbf{B}^T \mathbf{U}^T \mathbf{A}^T = \text{tr } \mathbf{U}^T \mathbf{A}^T \mathbf{B}^T \end{aligned}$$

thus

$$\frac{\partial \text{tr } \mathbf{A} \mathbf{X} \mathbf{B}}{\partial \mathbf{X}} = \mathbf{A}^T \mathbf{B}^T$$

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- We can generalise differentiation to vectors and matrices
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- When we look at **integration** it gets harder